

Attention-Enhanced U-Net with Transformer Encoder for Fetal Head

Segmentation and Growth Parameter Estimation

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21ECP401L-Major Project

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INTRODUCTION/ABSTRACT

- Fetal head biometry is essential for monitoring fetal growth and detecting abnormalities during pregnancy.
- Ultrasound imaging is widely used but manual measurement is time-consuming and operator dependent.
- This project uses a U-Net architecture to segment the fetal head from ultrasound images.
- The segmented mask is further used to compute key biometric parameters such as HC, BPD, OFD.
- Pixel-to-real-world conversion is applied using known pixel spacing to obtain measurements in millimeters.
- The predicted measurements are compared with ground truth values for performance evaluation.

MOTIVATION

- Accurate fetal head measurements such as HC, BPD, and OFD are critical for monitoring fetal growth and detecting abnormalities.
- Manual measurement from ultrasound images is time-consuming, operator-dependent, and prone to variability.
- Ultrasound images often suffer from noise, low contrast, and unclear boundaries, making precise segmentation difficult.
- Advances in Deep Learning and Convolutional Neural Networks provide an opportunity to automate and improve segmentation accuracy.
- Automated systems can ensure consistent, fast, and reproducible measurements, reducing human error.
- This project aims to bridge the gap between segmentation and real clinical parameter estimation, making the system more practical for real-world use.

OBJECTIVE

- To develop an automated system for fetal head segmentation from ultrasound images using deep learning.
- To implement a robust U-Net architecture for accurate mask prediction.
- To extract key fetal biometric parameters such as Head Circumference (HC), Biparietal Diameter (BPD), Occipitofrontal Diameter (OFD), and Head Area from the segmented mask.
- To convert pixel-based measurements into real-world units (millimeters) using pixel spacing information.
- To evaluate the accuracy of predicted measurements by comparing them with ground truth values.
- To design a fast, reliable, and clinically useful tool for automated fetal biometry.

BLOCK DIAGRAM

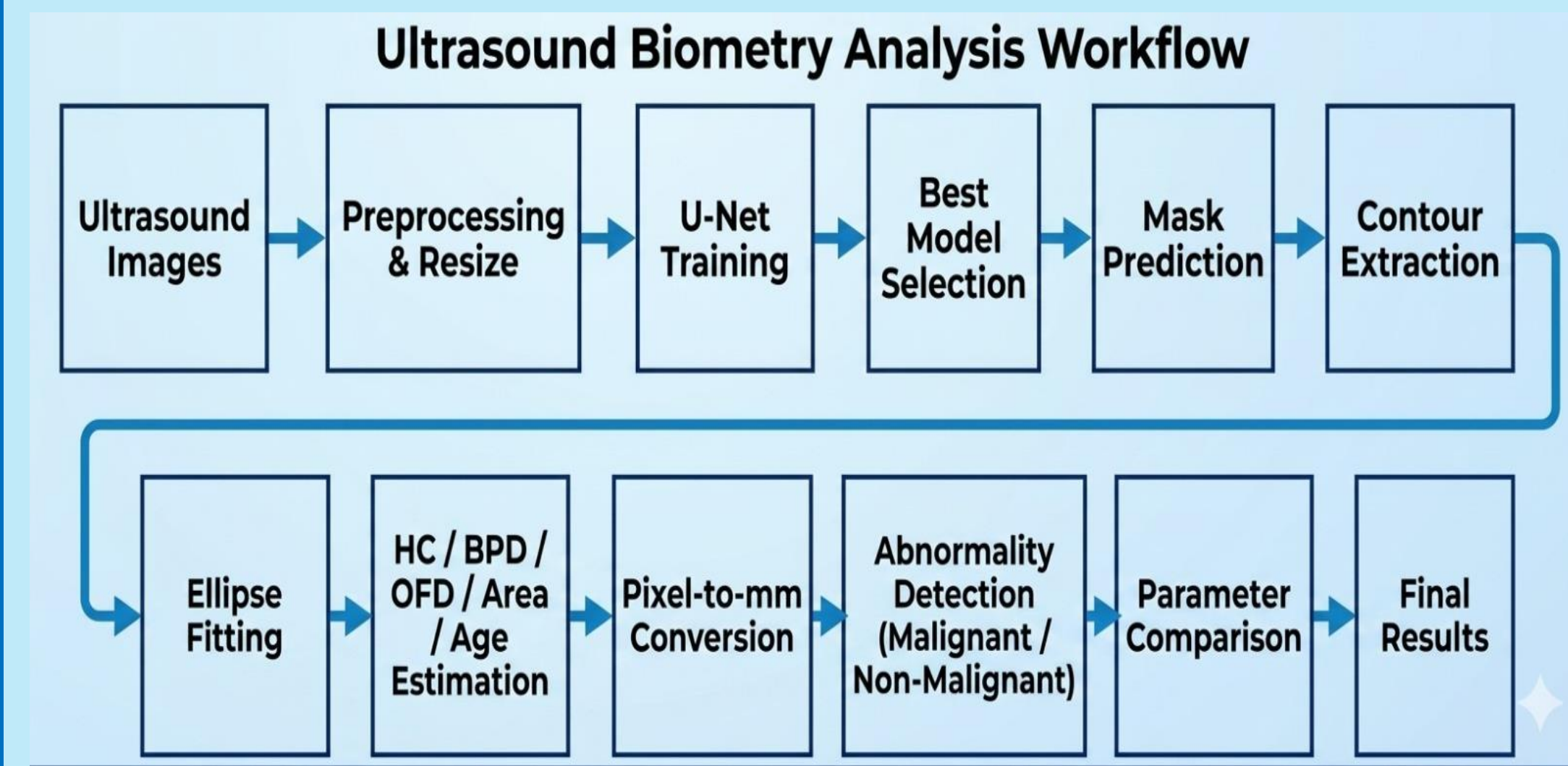


fig. 1. Block Diagram

DESIGN

- U-Net architecture is used to segment the fetal head from ultrasound images.
- Input images are resized and normalized to improve model performance and consistency.
- Data augmentation techniques are applied to enhance generalization and reduce overfitting.
- Ellipse fitting is applied to approximate the fetal head shape accurately.
- Key parameters such as HC, BPD, OFD, and Head Area are computed from the fitted ellipse.
- Pixel measurements are converted to millimeters using known pixel spacing values.
- Predicted measurements are compared with ground truth data for evaluation. Final output includes segmented masks, annotated images, and measurement results.

REALISTIC CONSTRAINTS

- Ultrasound images contain noise and low contrast, which can reduce segmentation accuracy.
- Variability in fetal position and scan conditions leads to inconsistent inputs.
- Limited annotated data affects the performance and generalization of Deep Learning models.
- Missing or inaccurate pixel spacing can impact the correctness of biometric measurements.

ENGINEERING STANDARDS

- IEEE 830 – Provides guidelines for clear and structured system design and documentation.
- ISO/IEC 25010 – Defines system quality in terms of performance, reliability, and efficiency.
- DICOM – Ensures standardized handling and exchange of medical imaging data.

RESULTS & INFERENCE

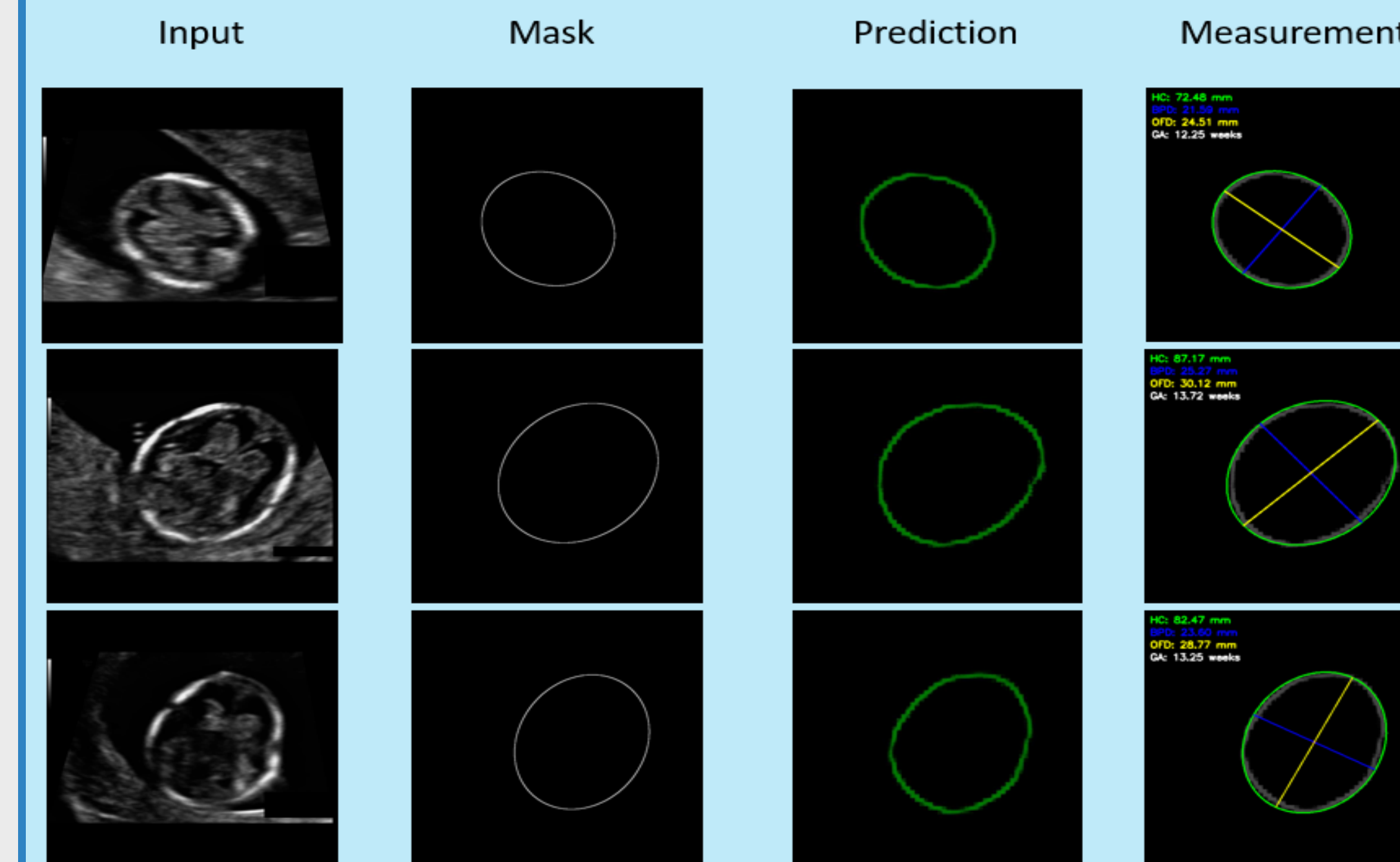


fig. 2. Segmentation result

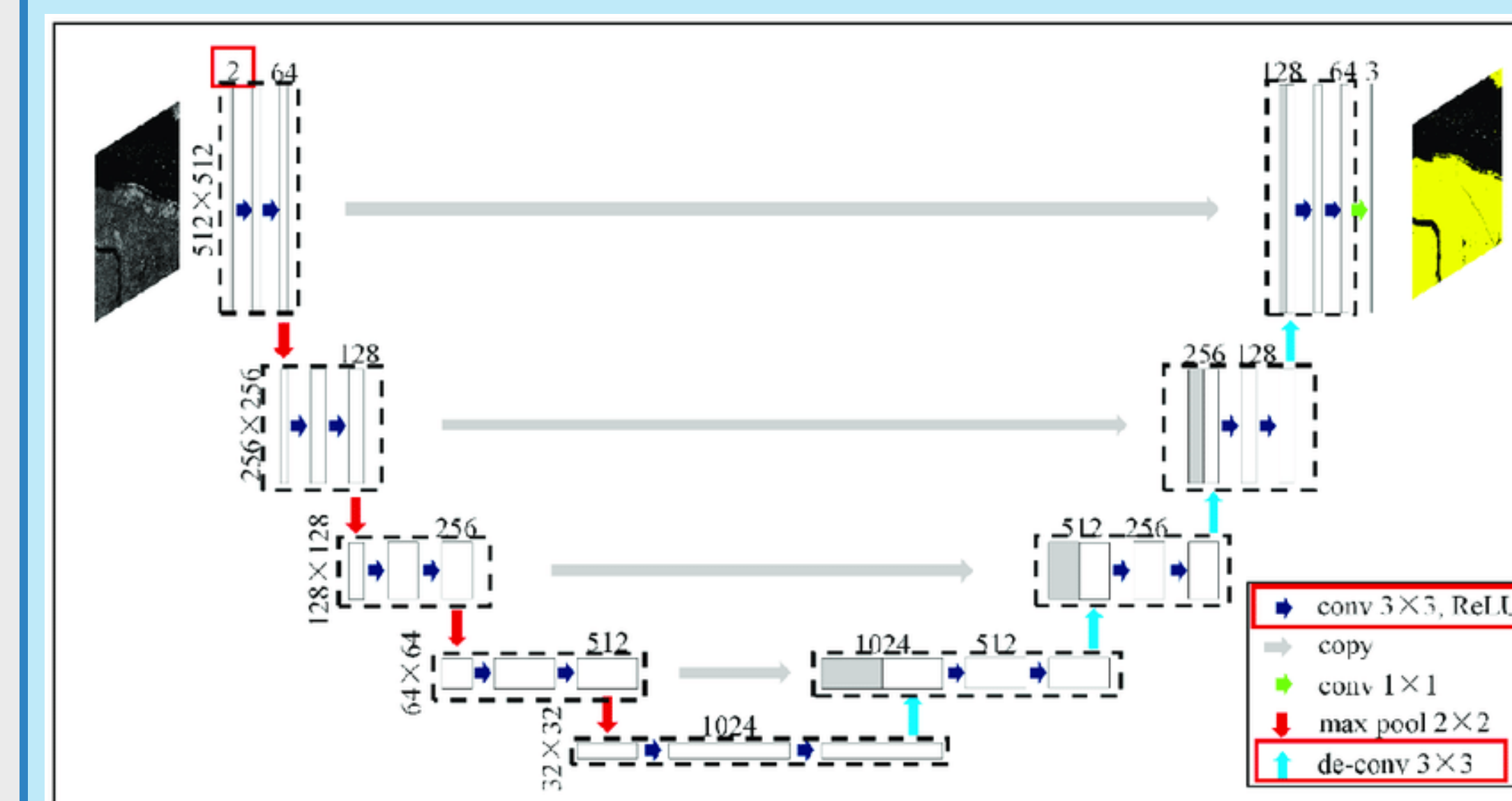


fig. 3. U-Net Architecture

- The proposed U-Net model accurately segments the fetal head region from ultrasound images, even in low contrast and noisy conditions.
- The predicted masks closely match the anatomical boundaries, ensuring precise and reliable segmentation.
- The encoder-decoder structure with skip connections enables effective feature extraction and preserves fine boundary details.
- Ellipse fitting is applied on the segmented region to compute key fetal parameters including Head Circumference (HC), Biparietal Diameter (BPD), Occipitofrontal Diameter (OFD), Head Area, and estimated gestational age.
- The extracted measurements are consistent and clinically relevant, demonstrating the effectiveness of combining deep learning with geometric analysis.
- The system maintains robustness across different image variations, handling shape and orientation differences effectively.
- A lightweight classification module is incorporated to indicate potential abnormal cases, supporting early-stage clinical assessment.

CONCLUSION

- An automated system for fetal head segmentation and biometry was successfully developed.
- The U-Net architecture provided accurate segmentation of ultrasound images.
- Key parameters such as HC, BPD, OFD, and Head Area were reliably computed from the predicted masks.
- The system achieved good agreement with ground truth measurements, demonstrating its effectiveness.
- This approach reduces manual effort and has potential for real-time clinical application in fetal monitoring.

REFERENCES

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SUSTAINABLE DEVELOPMENT GOALS

- SDG 3 – Good Health and Well-being
Enables early and accurate fetal health assessment through automated ultrasound analysis.
- SDG 9 – Industry, Innovation and Infrastructure
Promotes innovation by applying Artificial Intelligence in medical diagnostics.
- SDG 10 – Reduced Inequalities
Supports accessible and cost-effective healthcare solutions in resource-limited settings.

CONFERENCE /JOURNAL PUBLICATION

- Paper submitted to AIMLA 2026 International Conference
- Status: Under Review